

EXPERIMENT 8

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Question 1:

A company maintains customer information, product information, and sales order information in three separate tables: `customer_master`, `product_catalog`, and `sales_orders`.

The company wants to provide an order report to an external client. The report must contain the **Order ID, Order Date, Customer Name, Product Name, and Final Amount**.

For every order, first calculate the **Gross Amount** using:

Gross Amount = Unit Price × Quantity

Then calculate the **Discount Amount** using:

Discount Amount = Gross Amount × Discount Percentage / 100

Finally, calculate the **Final Amount** using:

Final Amount = Gross Amount – Discount Amount

Create a **complex view** using the three tables to generate this report. The view should expose only the required five columns and should not expose the customer's phone/email, product brand/unit price, or the order's quantity/discount percentage.

The company also wants to provide access to an external client. Create a PostgreSQL user named `client_user` and give this user **only SELECT permission on the view**. The client must not have direct access to the three original tables and must not be allowed to modify the view data.

Solution:

```
CREATE TABLE customer_master
```

```
(
```

```
    customer_id VARCHAR(5) PRIMARY KEY,
```

```
    full_name VARCHAR(50) NOT NULL,
```

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city VARCHAR(30)

);

CREATE TABLE product_catalog

(

product_id VARCHAR(5) PRIMARY KEY,

product_name VARCHAR(50) NOT NULL,

unit_price NUMERIC(10,2) NOT NULL

);

CREATE TABLE sales_orders

(

order_id SERIAL PRIMARY KEY,

product_id VARCHAR(5)

REFERENCES product_catalog(product_id),

quantity INT NOT NULL,

customer_id VARCHAR(5)

REFERENCES customer_master(customer_id),

discount_percent NUMERIC(5,2),

order_date DATE NOT NULL

);

INSERT INTO customer_master

(customer_id, full_name, city)

VALUES

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('C1', 'Amit Sharma', 'Delhi'),
('C2', 'Priya Verma', 'Mumbai'),
('C3', 'Ravi Kumar', 'Bangalore'),
('C4', 'Neha Singh', 'Kolkata'),
('C5', 'Arjun Mehta', 'Hyderabad');

```
INSERT INTO product_catalog
(product_id, product_name, unit_price)
VALUES
('P1', 'Smartphone X100', 25000),
('P2', 'Laptop Pro 15', 65000),
('P3', 'Wireless Earbuds', 5000),
('P4', 'Smartwatch Fit', 30000),
('P5', 'Gaming Console', 45000);
```

```
INSERT INTO sales_orders
(product_id, quantity, customer_id, discount_percent, order_date)
VALUES
('P1', 2, 'C1', 5, '2025-09-01'),
('P2', 1, 'C2', 10, '2025-09-02'),
('P3', 3, 'C3', 0, '2025-09-03'),
('P4', 1, 'C4', 8, '2025-09-04'),
('P5', 1, 'C5', 12, '2025-09-05'),
('P1', 1, 'C2', 5, '2025-09-06'),
('P3', 2, 'C1', 0, '2025-09-07');
```

--explain analyze

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create or replace view vW_FINAL_AMOUNT_CALCULATION

as

select o.order_id,o.order_date,c.full_name,p.product_name,

((p.unit_price*o.quantity)-((p.unit_price*o.quantity)*o.discount_percent/100)) as final_amount

from sales_orders as o

join customer_master as c on o.customer_id = c.customer_id

join product_catalog as p on p.product_id = o.product_id

select * from vW_FINAL_AMOUNT_CALCULATION;

create user CLIENT_123

PASSWORD '12345';

grant select on vW_FINAL_AMOUNT_CALCULATION to CLIENT_123

Output:

Data Output Messages Notifications					
Showing rows: 1 to 7 Page No: 1 of 1					
	order_id integer	order_date date	full_name character varying (50)	product_name character varying (50)	final_amount numeric
1	1	2025-09-01	Amit Sharma	Smartphone X100	47500.0000000000000000
2	2	2025-09-02	Priya Verma	Laptop Pro 15	58500.0000000000000000
3	3	2025-09-03	Ravi Kumar	Wireless Earbuds	15000.0000000000000000
4	4	2025-09-04	Neha Singh	Smartwatch Fit	27600.0000000000000000
5	5	2025-09-05	Arjun Mehta	Gaming Console	39600.0000000000000000
6	6	2025-09-06	Priya Verma	Smartphone X100	23750.0000000000000000
7	7	2025-09-07	Amit Sharma	Wireless Earbuds	10000.0000000000000000
Total rows: 7 Query complete 00:00:00.123 CRLF Ln 65, Col 1					

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Question 2:

An e-commerce company wants to regularly analyze the performance of its **product categories**.

The company stores product, category, customer, and order information in separate tables. To generate the category performance report, the database needs to perform multiple **JOINS**, **GROUP BY**, **aggregate functions**, **DISTINCT counting**, **calculated values**, and **CASE statements**.

Since the same report is generated frequently, executing the complete analytical query every time can increase database workload.

Create a PostgreSQL **Materialized View** named: mv_category_sales

Also demonstrate that when new order data is inserted, the Materialized View does **not automatically reflect the change** and must be refreshed.

Finally, use EXPLAIN ANALYZE to compare the performance of the original analytical query with the Materialized View query.

The report must contain:

- Category ID
- Category Name
- Total Products Sold
- Total Orders
- Unique Customers
- Total Revenue
- Average Revenue per Order
- Category Performance

Solution:

```
CREATE TABLE category_master (  
    category_id VARCHAR(5) PRIMARY KEY,  
    category_name VARCHAR(50) NOT NULL  
);
```

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```
CREATE TABLE product_master (  
    product_id VARCHAR(5) PRIMARY KEY,  
    product_name VARCHAR(50) NOT NULL,  
    category_id VARCHAR(5)  
        REFERENCES category_master(category_id)  
);
```

```
CREATE TABLE customer_master (  
    customer_id VARCHAR(5) PRIMARY KEY,  
    customer_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE order_details (  
    order_id VARCHAR(5) NOT NULL,  
    customer_id VARCHAR(5)  
        REFERENCES customer_master(customer_id),  
    product_id VARCHAR(5)  
        REFERENCES product_master(product_id),  
    quantity INT NOT NULL,  
    unit_price NUMERIC(10,2) NOT NULL  
);
```

```
INSERT INTO category_master  
(category_id, category_name)  
VALUES  
('C1', 'Electronics'),  
('C2', 'Clothing'),  
('C3', 'Books');
```

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```
INSERT INTO product_master
```

```
(product_id, product_name, category_id)
```

```
VALUES
```

```
('P101', 'Laptop', 'C1'),
```

```
('P102', 'Headphones', 'C1'),
```

```
('P201', 'Jacket', 'C2'),
```

```
('P202', 'Shoes', 'C2'),
```

```
('P301', 'SQL Book', 'C3');
```

```
INSERT INTO customer_master
```

```
(customer_id, customer_name)
```

```
VALUES
```

```
('U1', 'Aman'),
```

```
('U2', 'Priya'),
```

```
('U3', 'Rahul'),
```

```
('U4', 'Simran');
```

```
INSERT INTO order_details
```

```
(order_id, customer_id, product_id, quantity, unit_price)
```

```
VALUES
```

```
('O101', 'U1', 'P101', 1, 5000),
```

```
('O102', 'U2', 'P102', 2, 1000),
```

```
('O103', 'U1', 'P201', 1, 3000),
```

```
('O104', 'U3', 'P202', 2, 2000),
```

```
('O105', 'U4', 'P301', 3, 500),
```

```
('O106', 'U2', 'P101', 1, 5000),
```

```
('O107', 'U3', 'P201', 1, 3000),
```

```
('O108', 'U4', 'P102', 1, 1000);
```

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CREATE MATERIALIZED VIEW MV_CATEGORY_SALE

AS

SELECT

c.category_id,

c.category_name,

SUM(od.quantity) AS total_products_sold,

COUNT(DISTINCT od.order_id) AS total_orders,

COUNT(DISTINCT od.customer_id) AS unique_customers,

SUM(od.quantity * od.unit_price) AS total_revenue,

ROUND(

SUM(od.quantity * od.unit_price)

/ COUNT(DISTINCT od.order_id),

2

) AS avg_revenue_per_order,

CASE

WHEN SUM(od.quantity * od.unit_price) >= 10000

THEN 'Excellent'

WHEN SUM(od.quantity * od.unit_price) >= 6000

THEN 'Good'

WHEN SUM(od.quantity * od.unit_price) >= 3000

THEN 'Average'

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ELSE 'Low'

END AS category_performance

FROM category_master c

JOIN product_master p

ON c.category_id = p.category_id

JOIN order_details od

ON p.product_id = od.product_id

GROUP BY

c.category_id,

c.category_name;

EXPLAIN ANALYZE

SELECT * FROM MV_CATEGORY_SALE;

--REFRESH MV

REFRESH MATERIALIZED VIEW MV_CATEGORY_SALE;

Output:

Data Output Messages Notifications									
	category_id [PK] character varying (5)	category_name character varying (50)	total_products_sold bigint	total_orders bigint	unique_customers bigint	total_revenue numeric	avg_revenue_per_order numeric	category_performance text	
1	C1	Electronics	5	4	3	13000.00	3250.00	Excellent	
2	C2	Clothing	4	3	2	10000.00	3333.33	Excellent	
3	C3	Books	3	1	1	1500.00	1500.00	Low	